

BEYOND LOCAL ACTIVATION SPARSITY: MODEL-WIDE ZERO-PRODUCT OPPORTUNITY IN TRANSFORMERS

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ABSTRACT

Activation sparsity is often reported at one tensor, yet its computational meaning depends on where zero operands enter the entire transformer graph. We define *model-wide zero-product opportunity*, R_{model} : the fraction of scalar products in six declared transformer-block operations whose measured operands contain an exact zero, with a dense language-model head retained in the denominator. The metric includes post-RoPE query-key and causal probability-value products and is a logical count, not a runtime claim. We apply it to randomly initialized Pythia 14M, 70M, and 410M models trained for one matched 1.493B-token MiniPile pass. We compare evaluation-only clipping with two trained gate topologies using an Adam-aware orthogonal ℓ_1 pressure. The same local zero rate can imply very different R_{model} because operations have unequal footprints and zeros can be reused across attention products. At high gate threshold, the seven-site recipe reaches 27.5%, 40.6%, and 80.6% measured opportunity at 14M, 70M, and 410M, respectively. The 410M cohort also reverses the small-model trained loss response: stronger gating improves rather than degrades validation loss, even though matched post-hoc clipping still degrades it monotonically. A learning-rate screen does not repair its weak baseline, while its token exposure is only 3.7 tokens per parameter. We therefore treat 410M as a boundary result, not a scaling claim. These results support model-wide accounting as the comparison layer for sparse interventions and show why local sparsity, logical opportunity, and realized acceleration must remain separate quantities.

1 INTRODUCTION

The appeal of activation sparsity is immediate: if an operand is exactly zero, some multiply-accumulate work is mathematically unnecessary. The usual summary, however, is a zero percentage at a named tensor. That percentage is not a model-level computational quantity. A zero before a wide MLP projection has a different footprint from a zero in a query vector; a value zero can be reused over many causal queries; a pre-RoPE query zero need not remain zero after rotation; and a dense language-model head can dilute block sparsity. Consequently, two methods with similar local sparsity can expose very different amounts and locations of avoidable scalar products.

Prior work already recognizes this mismatch. TEAL aggregates per-matrix sparsity with matrix-footprint weights and applies training-free thresholding through linear layers (Liu et al., 2025). Our question is narrower and complementary: *given the actual operands of a declared full-sequence causal transformer graph, how many scalar products contain at least one exact zero?* We answer with R_{model} , an integer-counted metric spanning fused QKV, causal QK, causal PV, attention output, and both MLP projections, plus a dense LM-head denominator. This refinement matters wherever zeros are transformed or reused between named activations and multiplications. It is deliberately agnostic to kernel support and hardware speed.

Measurement also changes the experimental question. Rather than optimizing a single site’s sparsity, we compare validation loss against model-wide logical opportunity. We study post-hoc clipping and trained gates in Pythia models constructed from architecture configurations with random initialization (Biderman et al., 2023). The trained recipes place one-sided gates at the parallel attention and MLP

branch inputs and MLP hidden state, optionally adding post-RoPE query, key, value, and pre-output context gates. An Adam-aware orthogonal ℓ_1 update encourages small activations while removing the component that opposes the preconditioned task direction.

The results are not a simple “more sparsity is better” story. First, the operation-weighted composition is explanatory: the seven-site high-dose recipe adds 10–17 percentage points from QK and PV alone, and can exceed the four-site recipe despite lower zero mass at some local sites. Second, scale changes the observed trained response under our common-token protocol. At 14M and 70M, raising the gate threshold from 0 to 0.5 increases loss; at 410M it reduces loss by 0.50 and 0.31 for the four- and seven-site recipes. The evaluation-only curves do not reverse. Because all scales see the same 1.493B tokens, the 410M model receives far less exposure per parameter and has a weak baseline. A focused higher-learning-rate screen does not fix it, but longer training remains untested. We retain this cohort because the disagreement between trained and post-hoc behavior exposes an important boundary of the current evidence.

Our contributions are:

- an exact, pooled-integer definition of model-wide zero-product opportunity for a declared causal transformer workload, including operand placement and reuse that per-matrix summaries can miss;
- a joint accounting of quality, sitewise zero structure, and per-operation contributions for post-hoc and trained sparsification recipes;
- a matched-token Pythia case study through 410M that identifies a reproducible small-model pattern and a scientifically useful boundary result, while explicitly separating logical opportunity from runtime and scaling claims.

2 RELATED WORK

Sparse activations and training. Replacing smooth MLP nonlinearities with ReLU can yield high activation sparsity while preserving language-model quality (Mirzadeh et al., 2023). ProSparse stages regularization and thresholding during pretraining (Song et al., 2024); the Sparsing Law studies scaling behavior under sparsity-inducing training (Luo et al., 2024); and recent work examines sparsity-aware pretraining recipes at larger scales (Cetin et al., 2026). Conditional-computation and activation-sparsity studies likewise motivate training models whose dense weights are selectively exercised (Zhang et al., 2024). Our study is smaller and diagnostic: it holds a one-pass data budget fixed, reports exact product counts, and does not claim a compute-optimal sparse training recipe.

Post-training activation sparsification. TEAL selects activation thresholds without retraining and evaluates a model-level, matrix-footprint-weighted configuration (Liu et al., 2025). CATS and CHERS introduce contextual or structured activation selection (Lee et al., 2024; He et al., 2024), while R-Sparse combines rotations and thresholds to increase inference-time activation sparsity (Zhang et al., 2025). These works make efficiency concrete through kernels or latency measurements. We use a TEAL-like evaluation-only clipping family as a reference, but our central object is different: zero-containing scalar products reconstructed from actual operands, including causal attention reuse and a dense output head. Our metric is useful before kernel engineering, but cannot replace it.

Gradient interaction. Projecting conflicting auxiliary gradients has precedents in multi-task optimization, including PCGrad (Yu et al., 2020) and Adam-aware approaches such as Bloop (Hsieh et al., 2024). Our orthogonal ℓ_1 update uses the task optimizer’s second moment and a relative direction budget. It enforces a local non-opposition condition on the reconstructed directions; it does not guarantee that a finite parameter step improves task loss. The present comparison also does not isolate this optimizer from every topology and gate choice, so we treat it as part of a complete recipe rather than a standalone algorithmic contribution.

3 MODEL-WIDE ZERO-PRODUCT ACCOUNTING

Let a scalar product be a multiplication between two scalar operands within a declared matrix product. For operation o , let N_o be the number of such products in the evaluated workload and Z_o the number for which at least one *measured* operand is exactly zero. We count six operations in each Pythia

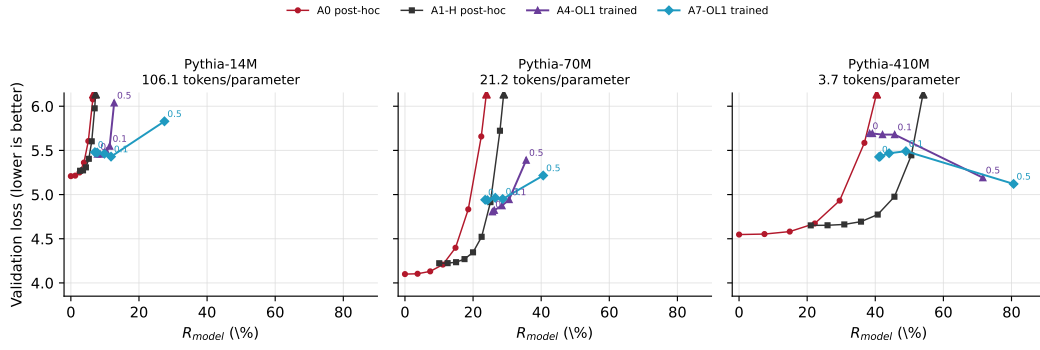


Figure 1: Validation loss versus measured model-wide zero-product opportunity. Lines connect gate thresholds (trained) or clipping targets (post-hoc); they are not fitted scaling curves. Upward triangles at the plot boundary mark post-hoc points whose losses exceed the displayed range. All points use one seed and complete validation.

block: fused QKV projection, causal QK scores, causal probability-value (PV) products, attention output projection, and the MLP up and down projections. Future positions are excluded from both QK and PV denominators. The language-model head is declared dense and contributes N_{head} only to the denominator:

$$R_{\text{model}} = \frac{\sum_{\ell=1}^L \sum_o Z_{\ell,o}}{\sum_{\ell=1}^L \sum_o N_{\ell,o} + N_{\text{head}}}. \quad (1)$$

Counts are accumulated as integers over all sequences and layers before the division. We use “model-wide” to mean this declared graph and workload, not all possible implementations of a language model.

For sequence length T , hidden width d , and MLP width d_f , the dense per-block denominator is

$$N_{\text{block}} = T(4d^2 + 2dd_f) + dT(T + 1), \quad (2)$$

where the final term combines the two valid-causal attention products. The model denominator is $LN_{\text{block}} + TdV$ for vocabulary size V . The metric is evaluated under eager attention so that the operands immediately before each product are observable.

This definition resolves three ambiguities in local sparsity. First, it weights zeros by exact operation footprints rather than by tensor size alone. Second, it counts reuse: one key or value coordinate can zero products at many valid query positions. Third, it places the observation after any intervening transformation. In particular, query and key are measured after partial RoPE; a zero before RoPE is not credited merely because it was once zero. Conversely, R_{model} does *not* estimate wall-clock acceleration. A dense kernel may execute every counted product, and sparse kernels face indexing, load balance, and shape-dependent overheads.

For descriptive context we also compute a topology-conditioned analytic reach $R_{\text{model,max}}$: the share of the same denominator belonging to operations that would become all-zero if the topology’s selected sites were all zero. It is neither an observed zero rate nor a universal upper bound on measured R_{model} , since the latter includes natural zeros at unselected operands. We therefore do not use its normalization as a primary result.

4 INTERVENTIONS

Sites and gates. Pythia uses parallel attention and MLP residual branches. We denote their separate LayerNorm outputs by a and m , the MLP nonlinear output by h , post-RoPE query and key by q and k , value by v , and concatenated attention context immediately before W_o by z . The one-sided and symmetric gates are

$$G_{\kappa}^{+}(x) = x \mathbf{1}[x \geq \kappa], \quad G_{\kappa}^{\pm}(x) = x \mathbf{1}[|x| \geq \kappa]. \quad (3)$$

Table 1: Matched one-pass exposure and A0 optimization context. Every model sees the same total tokens, not the same tokens per parameter. Validation loss is paired with the eager logical-product pass.

Model	Params	Tok./param.	Peak LR	Late train	Val.	Clip
14M	14.1M	106.1	1e-03	5.249	5.209	5/712
70M	70.4M	21.2	1e-03	4.009	4.100	8/712
410M	405.3M	3.7	3e-04	4.494	4.547	56/712

The comparison is detached: surviving values receive identity input gradient and rejected values zero gradient. At h , the gate replaces GELU.

The four-site recipe A4 gates and pressures $\{a, m, h, z\}$ with G_{κ}^{+} . The seven-site recipe A7 uses the same gates there and adds $G_{\kappa}^{\pm}$ at $\{q, k, v\}$. The shared gate threshold is $\kappa \in \{0, .01, .05, .1, .5\}$. Because the topologies change both gate and pressure sites, A4 versus A7 is a comparison between complete recipes, not a clean causal ablation of attention gating.

Orthogonal activation pressure. For captured tensors A_j , the auxiliary loss first averages within a tensor and then weights tensors equally: $L_1 = J^{-1} \sum_j \text{mean } |A_j|$. At each accumulated optimizer boundary we separately form the task and pressure gradients. AdamW consumes the globally clipped task gradient and updates its task-only moments. With the bias-corrected task moment $\hat{m}$ and task second moment $\hat{v}$, define

$$d_t = \frac{\hat{m}}{\sqrt{\hat{v} + \epsilon}}, \quad d_p = \frac{g_p}{\sqrt{\hat{v} + \epsilon}}. \quad (4)$$

If $\langle d_t, d_p \rangle < 0$, we remove the conflicting component from d_p ; otherwise it is unchanged. The resulting direction is scaled so $\lambda \|d_p\| / (\|d_t\| + \epsilon) \leq b$ and applied after AdamW with the same learning rate. We set pressure weight $\lambda = 1$ and relative budget $b = 1$. This “OL1” construction constrains the instantaneous preconditioned directions, not the finite-step loss.

Controls and post-hoc clipping. A0 retains stock GELU and no gates; A1-H replaces GELU with ReLU at h . From each trained control checkpoint, a TEAL-like evaluation-only sweep jointly clips $\{a, m, h, z\}$ at ten target sparsities from 0 to .9. Absolute-value thresholds are calibrated on the first ten complete training blocks, then held fixed for complete validation. These sweeps answer how a fixed model responds to removal; they are not trained sparse models.

5 EXPERIMENTAL PROTOCOL

Models, initialization, and data. We construct Pythia 14M, 70M, and 410M causal language models from pinned architecture configurations, never released weights. They have respectively 6/6/24 layers, hidden widths 128/512/1024, MLP widths 512/2048/4096, and 4/8/16 heads. Every condition within a scale strict-loads one seed-1234 FP32 initialization and consumes the same seed-1234 permutation. MiniPile (Kaddour, 2023) is tokenized with the pinned Pythia tokenizer, appending EOS per document and concatenating into non-overlapping 2,048-token blocks.

Each run has 712 optimizer boundaries, global batch 1,024 sequences, and 1,493,172,224 scheduled tokens: essentially one common MiniPile pass. This matches tokens, not optimization maturity. The resulting exposures are 106.1, 21.2, and 3.7 tokens per parameter (Table 1). All models use AdamW $(\beta_1, \beta_2) = (.9, .95)$, $\epsilon = 10^{-8}$, weight decay .1 excluding bias and normalization, global task-gradient clipping at 1, one-percent linear warmup, cosine decay to one tenth of peak, and zero dropout. Peak learning rate is 10^{-3} at 14M/70M and 3×10^{-4} at 410M. Training uses mixed precision with FP32 parameters and optimizer state.

Evaluation and diagnostics. Every canonical endpoint pairs cross-entropy loss and logical counts in the same eager forward pass over all 500 MiniPile validation documents: 338 complete sequences (692,224 input tokens), with the 1,444-token tail excluded. We pool exact-zero and near-zero counts at all seven intervention sites and the attention output, plus RMS and ℓ_2 accumulators. We also retain

Table 2: Low- and high-dose trained endpoints. Each cell is validation loss / R_{model} (%). A4-OL1 and A7-OL1 are complete recipes, not an isolated topology contrast.

Model	A4-OL1		A7-OL1	
	$\kappa = 0$	$\kappa = .5$	$\kappa = 0$	$\kappa = .5$
14M	5.458 / 7.8	6.038 / 12.7	5.480 / 7.1	5.829 / 27.5
70M	4.805 / 25.7	5.389 / 35.6	4.941 / 23.6	5.216 / 40.6
410M	5.692 / 38.3	5.191 / 71.6	5.426 / 41.1	5.121 / 80.6

per-layer weight norms, the six operation counts in Eq. 1, and boundary-level task/pressure gradient interaction. The post-hoc controls use identical validation coverage.

Scope. There is one seed, corpus, pass, and threshold ladder per scale. The study contains 30 trained endpoints (A0, A1-H, and ten A4/A7 conditions at each scale) and 60 post-hoc endpoints. Lines across them are evaluated paths; only nondominated points deserve the word Pareto frontier. Our primary comparisons are descriptive within scale. Cross-scale differences conflate architecture, exposure per parameter, and the selected peak learning rate.

6 RESULTS

6.1 QUALITY–OPPORTUNITY PATHS DIFFER BY INTERVENTION AND SCALE

Figure 1 shows all 90 evaluated points. The post-hoc A0 and A1-H paths initially buy opportunity at modest loss cost, then exit the plotted loss range as clipping approaches .9. The trained paths occupy different regions: their gates create substantial exact zeros already at $\kappa = 0$, while higher thresholds move them farther right. No one family dominates the entire evaluated set.

The endpoint ordering is consistent at high dose. At $\kappa = .5$, A7 has both lower loss and higher R_{model} than A4 at every scale (Table 2). Its opportunity advantage is large: 14.8, 5.0, and 9.0 percentage points at 14M, 70M, and 410M. At $\kappa = 0$, by contrast, A4 is better on both axes at 14M and 70M, whereas A7 is better at 410M. Thus the high-dose ordering replicates descriptively across the three cohorts; the zero-threshold ordering does not.

6.2 LOCAL ZERO MASS DOES NOT DETERMINE MODEL-WIDE OPPORTUNITY

Figure 2 resolves the low- and high-dose endpoints by activation site. One-sided gates yield broad exact-zero mass at a, m, h, z even when $\kappa = 0$. At $\kappa = .5$, A4 drives h and z to essentially 100% zeros, but leaves q, k, v dense. A7 instead creates large exact-zero mass throughout attention. Its high-dose (q, k, v) rates are (93.5, 94.5, 98.7)% at 14M, (71.5, 72.1, 86.9)% at 70M, and (77.8, 80.6, 94.0)% at 410M. This non-monotone cross-scale pattern already argues against a scalar “sparsity level” as the full explanation.

The operation decomposition in Figure 3 turns this local structure into its graph-level consequence. At 410M/ $\kappa = .5$, A4 has *more* zero mass than A7 at a (94.0 vs. 90.9%) and m (92.0 vs. 86.3%), yet A7 has higher R_{model} (80.6 vs. 71.6%). QK and PV contribute 11.1 percentage points available only to the seven-site recipe, overcoming its local deficit. Their combined contributions are 16.8, 10.4, and 11.1 points at 14M, 70M, and 410M. This is a direct example of topology and operand reuse changing a model-wide conclusion.

6.3 THE 410M COHORT IS A BOUNDARY RESULT

Within both trained 410M recipes, raising κ from 0 to .5 moves down and right: A4 changes by -0.501 loss and $+33.25$ opportunity points; A7 by -0.306 and $+39.49$ points. At 14M and 70M the corresponding loss changes are positive: $+0.580/ +0.584$ for A4 and $+0.349/ +0.275$ for A7. This is not a claim that high-dose sparse 410M beats its dense control—it does not. It is a reversal *within the trained threshold ladders*. The post-hoc paths at 410M continue to worsen monotonically,

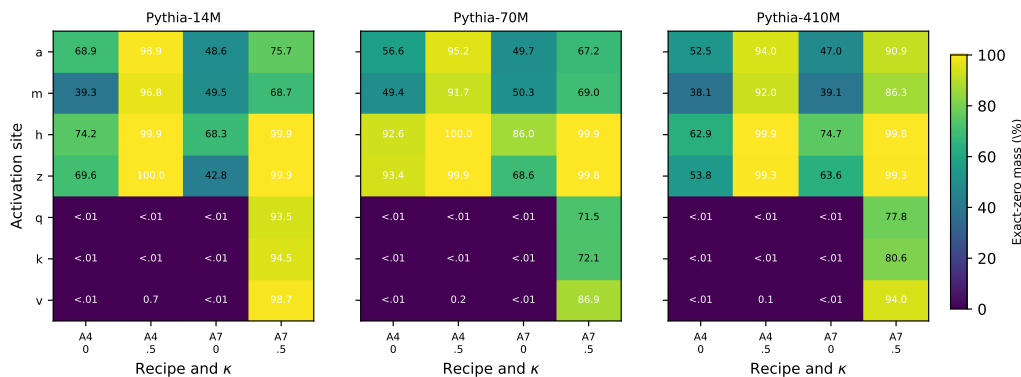


Figure 2: Integer-pooled exact-zero mass at seven activation sites for the trained low- and high-dose endpoints. q and k are measured after RoPE; columns are complete recipes. Values below .01% are printed as $<.01$.

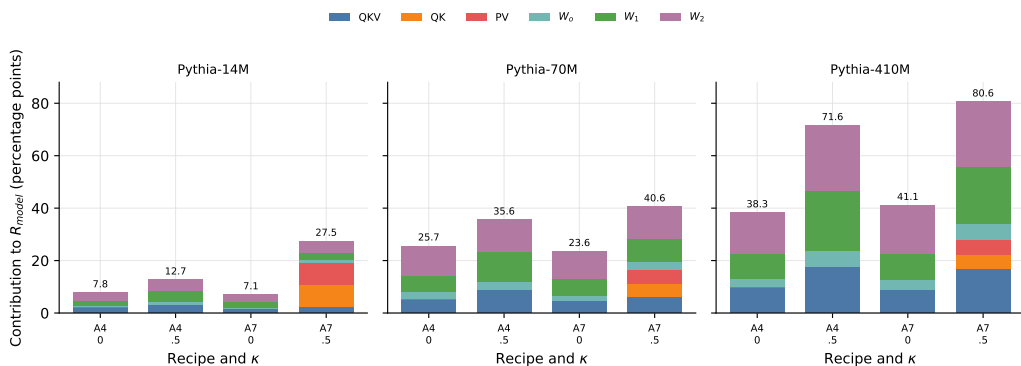


Figure 3: Contribution of each measured operation to R_{model} at trained endpoints. Segment heights use the common model denominator, so each stack sums to the labeled total. QK and PV use only valid causal positions.

separating an effect of training with the gates from robustness to evaluation-only deletion (Appendix Figure 4).

The shared-token design makes a scale-only interpretation untenable. Although 410M has six times the parameters of 70M, it receives the same tokens and only 3.7 tokens per parameter. Its A0 validation loss (4.547) is worse than 70M (4.100), it clips the global task gradient at 56 rather than 8 of 712 boundaries, and its maximum pre-clip norm is 26.96 rather than 3.51. These raw norms are not parameter-normalized, but they document a materially different optimization trajectory. A predeclared learning-rate screen selects the original 3×10^{-4} arm: mean late training loss is 4.494 at that rate, 4.513 at 6×10^{-4} , and 6.851 at 10^{-3} ; validation losses are 4.547, 4.552, and 6.787. The grid therefore rejects a simple upward-LR repair, while leaving training horizon unresolved.

7 DISCUSSION AND LIMITATIONS

What R_{model} adds. The metric is most useful as a common accounting layer. It makes the workload, operation set, causal mask, transformed operands, and dense head explicit; it then permits sitewise and operationwise diagnoses below the scalar summary. The A4/A7 comparison shows why this hierarchy matters: local zero percentages identify where representations collapse, while per-operation counts identify why the graph-level opportunity changes. A single number without those decompositions would hide the mechanism; local plots without the common denominator would not reveal its computational footprint.

How we use 410M. Excluding the cohort would produce a cleaner but misleading scale-persistence narrative. Including it as ordinary replication would overstate the design. We instead use it as a stress test: the high-dose topological advantage persists, but the trained loss response reverses under sharply lower exposure. The disagreement with post-hoc clipping suggests that learned adaptation to a gate can matter more than instantaneous deletion. It does not distinguish beneficial regularization, altered optimization geometry, capacity effects, or undertraining. A matched tokens-per-parameter or loss-matched study and replicate seeds are required before any scaling statement.

Sitewise redistribution is selective. The cross-scale endpoint heatmap shows that broader pressure does not induce a uniform attention response. An earlier matched 14M diagnostic reinforces this point: as near-zero mass at directly pressured h rose from 63.5% to 92.4%, v moved only from .133% to .185%; q , k , and m remained tiny and non-monotone, while the post- W_o attention output rose through the middle doses and then fell. We treat “spillover” as this site-specific descriptive redistribution, not a causal propagation mechanism (Appendix Figure 6).

Limits of the intervention evidence. Each endpoint has one seed. A4 and A7 differ jointly in gates and pressure sites. The pressure weight and trust budget were not broadly tuned. A matched 14M comparison of naive and orthogonal ℓ_1 pressure produced small, mixed loss differences, so the present data do not establish OL1 as superior (Appendix E). Post-hoc thresholds are calibrated on ten training blocks, not fit to validation, but they remain a particular clipping rule rather than a universal comparator.

Limits of the computational claim. Our denominator describes complete length-2,048 sequences with uncached causal attention. It resembles prefill or training accounting, not cached autoregressive decoding, whose KV reuse changes operation footprints. Exact zeros alone do not create sparse execution. We report no removed FLOPs, latency, energy, memory-traffic benefit, or end-to-end speedup. Establishing those outcomes requires kernels whose format and scheduling match the measured site structure.

8 CONCLUSION

Activation sparsity becomes scientifically comparable only after declaring where zeros meet products. R_{model} counts that opportunity across the causal transformer graph while keeping local sites and operation contributions inspectable. Across three Pythia scales, broad attention gating can create substantially more logical opportunity than highly sparse MLP-adjacent sites, but quality responses depend on intervention and training regime. The 410M reversal prevents a premature scaling story and sharpens the next empirical question: whether trained quality–opportunity behavior persists under matched optimization maturity. Until sparse kernels are measured, the result remains a map of exact-zero products, not a speed claim.

REPRODUCIBILITY STATEMENT

The appendix specifies the model, data, initialization, optimization, gate, pressure, coverage, and count-aggregation contracts. The supplementary material includes machine-readable source identities, a claim ledger, figure observations, and a clean-repository handoff. Every reported validation point covers the same 338 complete sequences, and every reported fraction is rebuilt from pooled integer counts. Code and immutable artifact manifests will be released with the submission.

AI USE DISCLOSURE

Generative AI tools assisted with methodological criticism, literature search, implementation and test review, figure code, result interpretation, and prose drafting and editing. The authors selected the research questions and experimental designs and are responsible for independently checking every claim, citation, computation, and artifact before submission.

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A OPERATIONAL DETAILS

A.1 ACTIVATION PORTS

Table 3 fixes each symbol to the tensor consumed by a downstream product. Pythia’s a and m are distinct LayerNorm outputs on parallel residual branches. “Attention output” in the diagnostics is after W_o and is distinct from z .

Table 3: Canonical activation ports.

Site	Exact port	Shape	Downstream product
a	attention-branch norm/gate output	$[B, T, d]$	fused QKV
m	MLP-branch norm/gate output	$[B, T, d]$	MLP up/ W_1
h	MLP nonlinearity/gate output	$[B, T, d_f]$	MLP down/ W_2
q, k	immediately after partial RoPE	$[B, H, T, d_h]$	causal QK
v	value from fused QKV	$[B, H, T, d_h]$	causal PV
z	concatenated context before W_o	$[B, T, d]$	attention output/ W_o

For QK, a coordinate product at valid pair (i, j) is counted zero if $q_{i,r} = 0$ or $k_{j,r} = 0$. For PV, it is counted zero if the causal attention probability $p_{i,j} = 0$ or $v_{j,r} = 0$. Masked future pairs are absent, rather than credited as zeros. Projection products use the corresponding activation and weight operands. Exact zero means floating-point equality to zero; the 10^{-3} and 10^{-2} near-zero diagnostics are separate metrics.

A.2 TOPOLOGY-CONDITIONED ANALYTIC REACH

Table 4: Analytic selected-site reach $R_{\text{model},\text{max}}$ for full length-2,048 sequences. It is not an observed sparsity or speedup.

Topology	Pythia-14M	Pythia-70M	Pythia-410M
A4 (a, m, h, z)	12.833%	37.063%	74.776%
A7 (a, m, h, q, k, v, z)	29.952%	49.424%	87.245%

The reach numerator credits a selected site to the operation it can null: $a \rightarrow \text{QKV}$, $m \rightarrow W_1$, $h \rightarrow W_2$, q or $k \rightarrow \text{QK}$ once, $v \rightarrow \text{PV}$ and (if all values vanish) W_o , and $z \rightarrow W_o$. The denominator is exactly that of Eq. 1. Since measured R_{model} counts natural zeros everywhere, it is not required to lie below this selected-site analytic quantity.

B ADDITIONAL QUALITY-OPPORTUNITY VIEWS

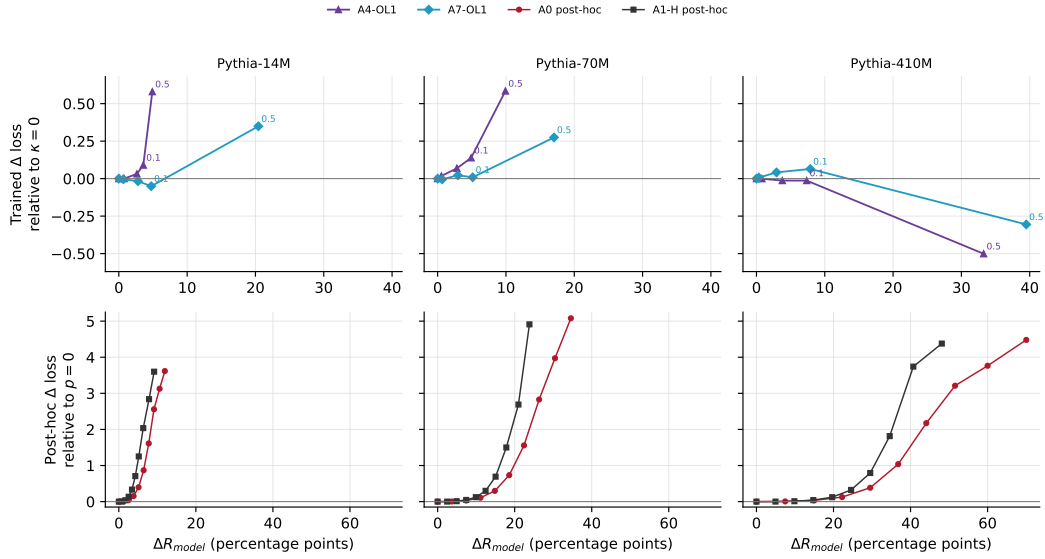


Figure 4: Within-family change relative to each zero-dose point. Top: trained threshold ladders. Bottom: evaluation-only clipping paths. The trained 410M direction reverses while both post-hoc families retain positive loss cost.

Table 5: Pythia-410M A0 learning-rate screen. Selection was predeclared on the mean task loss over boundaries 649–712, not validation loss.

Peak LR	Late train loss	Validation loss	Selected
3×10^{-4}	4.493717	4.547456	yes
6×10^{-4}	4.513309	4.552068	no
1×10^{-3}	6.851160	6.786663	no

C TRAINING DYNAMICS

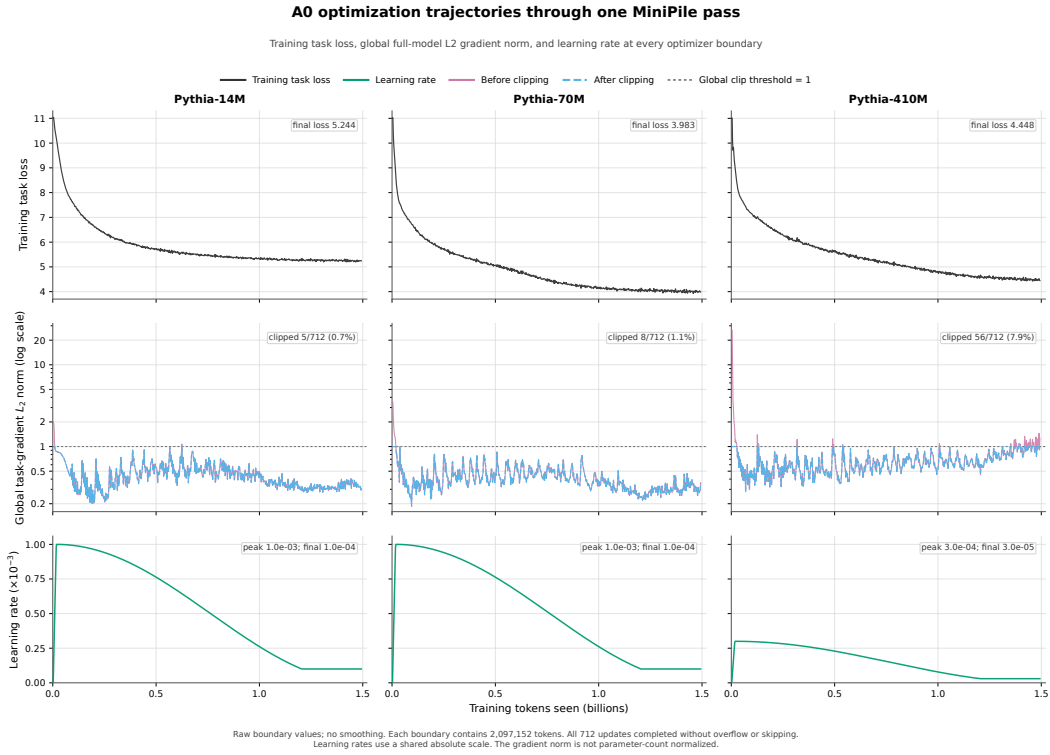


Figure 5: A0 training loss, global task-gradient norm before and after clipping, and learning rate over the matched token budget. These are unnormalized global norms, so cross-scale magnitude is descriptive rather than a parameter-size-controlled comparison.

All streams contain 712 finite boundaries and no skipped update. A0 has 5/8/56 clipped boundaries and maximum pre-clip norms 2.002/3.513/26.962 at 14M/70M/410M. The figure is included to expose the executed optimization regime, not to diagnose the 410M reversal from one scalar trace.

D EXPLORATORY SITEWISE REDISTRIBUTION

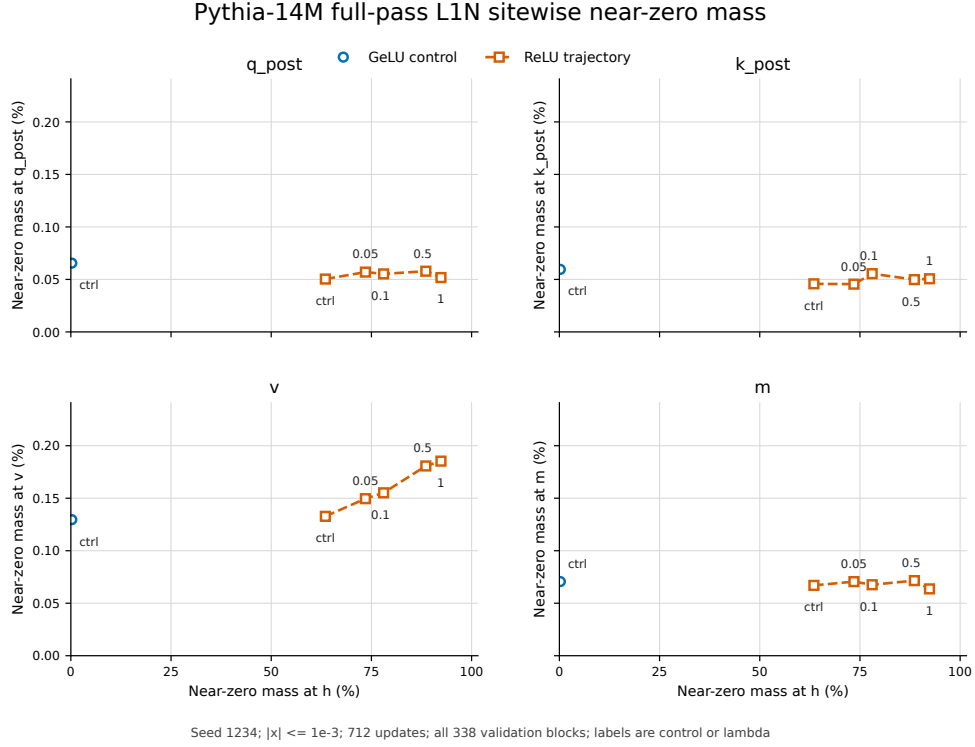


Figure 6: Pooled near-zero mass ($|x| \leq 10^{-3}$) at unpressured sites versus directly pressured h in a separate Pythia-14M ReLU/naive- ℓ_1 cohort. Lines order pressure dose and are not temporal trajectories.

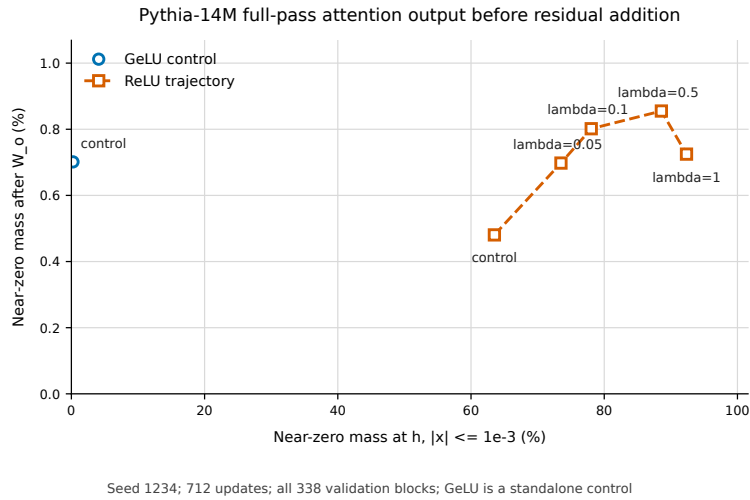


Figure 7: Near-zero mass after W_o versus at h in the same exploratory cohort. The output response rises through the middle doses and falls at the largest pressure.

All points in Figures 6–7 cover the same complete validation set. They use near-zero rather than exact-zero mass and a different pressure recipe from the main ladder; they support only the descriptive statement that redistribution is site-specific and non-monotone.

E ORTHOGONAL-PRESSURE DIAGNOSTIC

In a matched Pythia-14M A1-H cohort, OL1 minus naive- ℓ_1 validation loss was -0.0081 and -0.0061 at $\lambda = .05$ and $.1$, then -0.0025 and $+0.0188$ at $\lambda = .5$ and 1 . Exact-zero mass and R_{model} were nearly matched. Along naive- ℓ_1 trajectories, negative raw task–pressure dot products occurred at 49.2–70.9% of boundaries as dose increased. OL1 projected its own negative Adam-relative interactions to the orthogonal boundary, but this geometric guarantee did not yield a consistently better endpoint.

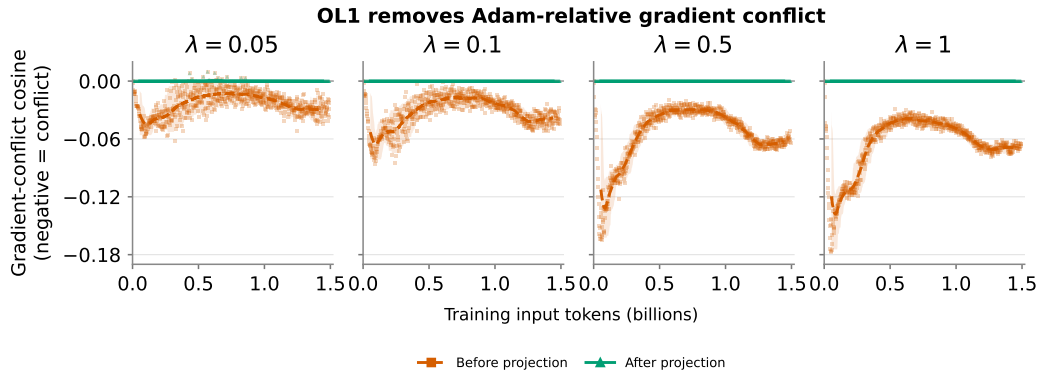


Figure 8: OL1 Adam-relative task–pressure interaction before and after the operational projection in the separate 14M diagnostic. Rolling summaries describe temporally dependent boundaries, not replicate uncertainty.

F REPRODUCIBILITY IDENTITIES

The dataset is JeanKaddour/minipile at revision

18ad1b0c701eaa0de03d3cecfdd769cbc70ffbd0.

The tokenizer is EleutherAI/pythia-14m-deduped at revision

7386d9a4ae45aef494a6e704910394def3037fc5.

Encoding adds no special tokens and appends EOS per document. The training cache contains 1,491,711,416 tokens and 728,374 complete blocks (tail 1,464); the validation cache contains 693,668 tokens and 338 complete blocks (tail 1,444). Their SHA-256 values are:

Training cache:

da82a2ea2e0080c7fd681c7a93b07d3d9ff3d5357a8640895a82d536a1eaf97c

Validation cache:

51cd758fda72f14383da30c358a895d0223c0d1d80b31455d2d842c3656d0451.

The 410M cohort uses 24 layers, $d = 1024$, $d_f = 4096$, 16 heads, vocabulary 50,304, and 405,334,016 parameters. Its canonical initialization has parameter SHA-256

76217bf2ef13de2377c7750515a559cb71f574c274476e08408a5f7749ea9cff.

Mixed GPU models were used for this cohort; condition assignment is execution provenance and not an intended scientific factor. Each retrieved archive was hash-verified before cloud teardown.